■ Java 100-Question Foundation Masterclass

Objective: Master Java fundamentals through progressive, learn-by-doing challenges. Follow each module in order and implement every problem yourself for maximum learning.

■ Module 1: Logic & Arithmetic (1–20)

1. Digit Sum – Find the sum of digits of a number.
2. Reverse Integer – Reverse a number without using strings.
3. Palindrome Number – Check whether a number reads the same backward.
4. Armstrong Number – Validate if sum of digit powers equals the number.
5. Prime Check – Determine whether a number is prime.
6. Fibonacci Series – Print first N Fibonacci numbers.
7. GCD & LCM – Compute greatest common divisor and least common multiple.
8. Strong Number – Sum of factorial of digits equals the number.
9. Decimal to Binary – Convert decimal to binary manually.
10. Factorial – Compute N! using loops.
11. Harshad Number – Number divisible by sum of digits.
12. Perfect Number – Sum of divisors equals the number.
13. Spy Number – Digit sum equals digit product.
14. Automorphic Number – Square ends with the number.
15. Neon Number – Digit sum of square equals the number.
16. Prime Range – Print primes between X and Y.
17. Power Logic – Compute X^Y without Math.pow().
18. Leap Year – Validate leap year using conditions.
19. Simple Interest – Calculate SI from P, R, T.
20. Menu Calculator – Build calculator using switch-case.

■ Module 2: Array Mastery (21–40)

1. Second Largest – Find the second highest element.
2. Left Rotation – Rotate array elements left.
3. Frequency Count – Count occurrence of elements.
4. Reverse Array – Reverse without extra array.
5. Remove Duplicates – From sorted array.

6. Merge Arrays – Combine two arrays.
7. Missing Number – Find missing value from 1 to N.
8. Even-Odd Sort – Evens first, odds later.
9. Linear Search – Find element position.
10. Binary Search – Divide and conquer search.
11. Matrix Sum – Add two 2x2 matrices.
12. Transpose – Convert rows to columns.
13. Diagonal Sum – Sum main diagonal elements.
14. Min-Max 2D – Find min & max in matrix.
15. Array Equality – Compare two arrays.
16. Bubble Sort – Implement sorting algorithm.
17. Selection Sort – Implement selection sort.
18. Sub-array Sum – Check subarray equals K.
19. Common Elements – Intersection of arrays.
20. Matrix Multiplication – Multiply 3x3 matrices.

■ Module 3: Strings & Text (41–60)

1. Reverse String – Without reverse() method.
2. Vowel/Consonant Count – Count letters.
3. Word Count – Count words in sentence.
4. Anagram Check – Validate two strings.
5. Palindrome String – Check reversed equality.
6. Manual Case Conversion – A to a without methods.
7. Toggle Case – Swap character cases.
8. First Unique Character – Non-repeating char.
9. Substring Logic – Manual contains logic.
10. String Compression – aaabb → a3b2.
11. Longest Word – In a sentence.
12. Strip Spaces – Remove all spaces.
13. Category Count – Letters, digits, symbols.
14. Rotation Check – Detect rotated strings.
15. Replace Character – Replace e with *.
16. Performance Test – String vs StringBuilder.

17. Domain Extract – From email address.
18. Initials – Convert full name to initials.
19. ASCII Sum – Sum ASCII values.
20. Alphabetical Sort – Sort string characters.

■ Module 4: OOP Foundation (61–80)

1. Basic Class – Create Car class.
2. Constructors – Initialize Employee data.
3. Method Overloading – Area calculation.
4. this Keyword – Resolve naming conflict.
5. Static Counter – Count objects created.
6. Static Method – Unit conversion method.
7. Encapsulation – Private fields with getters/setters.
8. Inheritance – Animal to Cat.
9. Multi-level Inheritance – Vehicle hierarchy.
10. Method Overriding – Polymorphic behavior.
11. super Keyword – Call parent constructor.
12. Final Variable – Demonstrate immutability.
13. Abstract Class – Bank interest method.
14. Interface – Playable implementation.
15. Polymorphism – List of shapes.
16. Garbage Collection – finalize() tracking.
17. Aggregation – Department has Teacher.
18. Overloaded Methods – Add operations.
19. Access Levels – public vs private.
20. Inner Class – Battery inside Laptop.

■ Module 5: Practical Tools & Projects (81–100)

1. ArrayList Operations – Add & remove names.
2. Sort List – Sort integers.
3. HashMap – Phonebook implementation.
4. HashSet – Unique elements extraction.
5. Stack – Push and pop elements.

6. Queue – FIFO waitlist.
7. Iterator – Remove elements safely.
8. Arithmetic Exception – Handle division by zero.
9. NullPointerException – Safe null handling.
10. Custom Exception – InvalidAgeException.
11. Write File – Save text to file.
12. Read File – Read file contents.
13. Date Format – Print current date.
14. Random Dice – Generate 1–6.
15. Login Logic – 3 attempt system.
16. GPA Calculator – Convert marks.
17. Unit Converter – KG to LBS.
18. Shopping Cart – Sum product prices.
19. Lambda Expressions – Filter even numbers.
20. Capstone – Student Management System.